

Let Σ and Γ be finite alphabet sets.

Homomorphism $h: \Sigma^* \rightarrow \Gamma^*$ such that $\forall x, y \in \Sigma^*$

$$h(xy) = h(x)h(y) \quad \text{--- (H1)}$$

$$h(\epsilon) = \epsilon \quad \text{--- (H2)}$$

Any homomorphism defined on Σ^* is uniquely determined by its values on Σ .

$$\Sigma = \{a, b\} \quad \Gamma = \{c, d, e\} \quad h(a) = cde, h(b) = dd$$

$$h(aab) = cde cde dd = h(a)h(a)h(b)$$

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$$\text{For } A \subseteq \Sigma^* \text{ let } h(A) = \{h(x) \mid x \in A\} \subseteq \Gamma^*$$

$$\text{For } B \subseteq \Gamma^* \text{ let } h^{-1}(B) = \{x \mid h(x) \in B\} \subseteq \Sigma^*$$

$h(A)$: image of A under h .

$h^{-1}(B)$: preimage of B under h .

Theorem 1. Let $h: \Sigma^* \rightarrow \Gamma^*$ be a homomorphism
if $A \subseteq \Sigma^*$ is regular then $h(A)$ is regular

Theorem 2. Let $h: \Sigma^* \rightarrow \Gamma^*$ be a homomorphism.

if $B \subseteq \Gamma^*$ is regular then $h^{-1}(B)$ is regular.

An application

Definition. NFA with ϵ -transitions

$$M = (Q, \Sigma, \epsilon, \Delta, S, F)$$

Such that ϵ is a special symbol not in Σ .

Let $M_\epsilon = (Q, \Sigma \cup \{\epsilon\}, \Delta, S, F)$ - ordinary NFA.
over $\Sigma \cup \{\epsilon\}$.

Definition. Acceptance for NFA with ϵ -transition.

M accepts $x \in \Sigma^*$ if $\exists y \in (\Sigma \cup \{\epsilon\})^*$ s.t

- M_ϵ accepts y (under the ordinary definition of acceptance for NFA)
- x is obtained from y by erasing all occurrence of symbol ϵ .

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of symbol ϵ .

That is: $x = h(y)$ where $h: (\Sigma \cup \{\epsilon\})^* \rightarrow \Sigma^*$
is the homomorphism defined as:

$$h(a) = a, \text{ for } a \in \Sigma$$

$$h(\epsilon) = \epsilon$$

$$\text{Thus } L(M) = h(L(M_\epsilon))$$

Theorem 2. Let $h: \Sigma^* \rightarrow \Gamma^*$ be a homomorphism.

if $B \subseteq \Gamma^*$ is regular then $h^{-1}(B)$ is regular.

Proof of Theorem 2. Let $M = (Q, \Gamma, \delta, s, F)$ - DFA

Such that $L(M) = B$. To show: $\exists M'$ over Σ s.t.
 $L(M') = h^{-1}(B)$.

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Definition of M' : $M' = (Q, \Sigma, \delta', \delta_0, F)$

$$\delta'(q, a) = \delta(q, \underbrace{h(a)}_{\in \Gamma^*})$$

$\downarrow$
 $a \in \Sigma$

Lemma 1. $\hat{\delta}'(q, x) = \hat{\delta}(q, h(x))$.

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 $L(M') = h^{-1}(B)$.

Definition of M' : $M' = (Q, \Sigma, \delta', s, F)$

$$\begin{array}{ccc} \delta'(q, a) & = & \hat{\delta}(q, h(a)) \\ \downarrow & & \downarrow \\ \in \Sigma & & \in \Gamma^* \end{array}$$

Lemma 1. $\hat{\delta}'(q, x) = \hat{\delta}(q, h(x))$.

Proof.

Induction on $|x|$. Base case $x = \epsilon$ is trivial.

$$\begin{aligned} \hat{\delta}'(q, xa) &= \delta'(\hat{\delta}'(q, x), a) \quad [\text{Def of } \hat{\delta}'] \\ &= \delta'(\hat{\delta}(q, h(x)), a) \quad [\text{I.H.}] \\ &= \hat{\delta}(\hat{\delta}(q, h(x)), h(a)) \quad [\text{Def of } \delta'] \\ &= \hat{\delta}(q, h(x)h(a)) \\ &= \hat{\delta}(q, h(xa)) \quad [\text{property (H1)}] \end{aligned}$$

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To show $L(M') = h^{-1}(B)$

It suffices to show $L(M') = h^{-1}(L(M))$

For any $x \in \Sigma^*$,

$x \in L(M')$ iff $\hat{\delta}'(q, x) \in F$ [Defn of acceptance]

iff $\hat{\delta}(q, h(x)) \in F$ [Lemma 1]

iff $h(x) \in L(M)$ [Defn. of acceptance]

iff $x \in h^{-1}(L(M))$ [Defn. of $h^{-1}(L(M))$]